

EMILIA MICHALEK

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EDUCATION

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN (UofI)

Graduation: May 2027

Bachelor of Science in Computer Science

GPA: 3.97/4.00

Master of Computer Science (BS-MCS Combined Program)

Relevant Coursework: Intro to Algs & Models of Comp, Numerical Methods (Python), Systems Programming (C), Database Systems (MYSQL), Applied ML (Python), Artificial Intelligence (Python), Computer Security I (C, html, etc.)

OAKTON COMMUNITY COLLEGE, DES PLAINES, IL

Aug 2022 – May 2024

Engineering Pathways Program Transfer

GPA: 3.69/4.00

Relevant Coursework: C++ Computer Science I, C++ Data Structures, C++ Objects & Algorithms, Python Computer Science I, Computer Architecture & Organization, Discrete Mathematics

EXPERIENCE

SOFTWARE ENGINEERING INTERN AT MOTOROLA SOLUTIONS, RESEARCH PARK

Mar 2025 – Present

Developing production-quality software and analytics solutions for customer-facing and internal applications

- Built an end-to-end internal analytics solution, implementing parsing/decryption of new device fields, integrating them into an existing ETL pipeline, developing a Power BI dashboard, adding unit-test coverage, and deploying the solution to pre-production.
- Partnered with stakeholder and distributed engineering teams to define requirements, scope future enhancements, and deliver RadioCentral features used by thousands of customers, reducing clicks through image previews and radio-ID autofill.

COURSE ASSISTANT FOR CS 211/210 (UofI)

Spring 2025; Spring 2026

Weekly Assignment Grader for Ethical and Professional Conduct

- Ensure understanding and application of core concepts in roughly 20 student papers a week
- Give comprehensive feedback on strengths and weaknesses of essay structure and concept understanding

GEARUP RESEARCH PROGRAM (UofI)

Jun 2024 – Jan 2025

Detecting Spoofing and Jamming on Android Devices

- Analyzed data collected by existing tools and applications to inform the development of a comprehensive application for detecting spoofing & jamming using only limited sensors available on Android devices
- Took advantage of Doppler shift's effect on signal waves sent from satellite to aid in creating alternative solution to detecting spoofing and jamming without requiring specialized equipment
- Collaborated with small team to research, develop, and/or organize tools and software necessary to test theories

PROJECTS

STOCK INVESTMENT WEB APPLICATION

Jun 2024 – Jul 2024

- Collaborated with team of four classmates to create functional web application capable of pulling live stock prices from yfinance API utilizing MySQL back end to organize and access user account and portfolio information
- Developed code and queries in MySQL, Python (Flask), HTML, and CSS

RSA AND HUFFMAN ENCRYPTION ALGORITHMS (C++)

Nov 2022 – Dec 2023

LEADERSHIP AND ACTIVITIES

AIR FORCE ROTC PARTICIPANT (UofI)

Aug 2024 – Dec 2024

- Coordinated with team of cadets to complete tasks with efficiency, cohesion, and precision
- Cultivated foundational skills in leadership, discipline, and strategic decision-making

SKILLS

TECHNICAL: Python, C++, C, Java, MySQL, JavaScript, Typescript, HTML, CSS, PowerBI

LANGUAGES: English (Fluent), Polish (Fluent)